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Editor-in-Chief
IEEE Access

Re: Submission of “Mechanically Verified Optimization and UPPAAL Export for Timed Büchi Automata”

Dear Editor-in-Chief,

On behalf of my coauthors, I submit “Mechanically Verified Optimization and UPPAAL Export for Timed Büchi Automata” for consideration by *IEEE Access*. The paper presents Rocq-verified, language-preserving optimization and UPPAAL-oriented export transformations for timed Büchi automata. It proves generic preservation results for arbitrary input automata and any finite number of optimization rounds, extracts the verified transformations to OCaml, and evaluates the integrated implementation on 31 benchmark configurations and four system-model case studies.

We are also submitting a distinct companion manuscript, “A Mechanically Verified Translation of $MTL_{0,\infty}$ into Timed Büchi Automata,” separately to IEEE Transactions on Software Engineering. Both manuscripts contain the same upstream Rocq modules for the shared translation and its correctness proof. The TSE manuscript presents and mechanizes the MTL-to-TBA translation theorem, including completeness under overlapping activations. This IEEE Access paper reuses that theorem as an upstream prerequisite and does not claim it as a new result. Its distinct contributions are the generic TBA optimization and export proofs, their extracted implementation, and the evaluation described above.

This manuscript is original, has not been published, and is not under consideration by another venue. It is a distinct article from the separately submitted TSE companion manuscript; the shared proof material is disclosed above rather than claimed as two independent results. The complete companion manuscript is supplied as related material for editorial assessment.

Thank you for considering our submission.

Sincerely,

Elie Fares

for the coauthors: Jean-Paul Bodeveix, Hussain Al-Aqrabi, and Azar Salami